Code No: 117HN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, November/December - 2017
SOFT COMPUTING
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Define best first search. [2]
b) Discuss briefly about means end analysis. [3]
c) Define associative memory networks. [2]
d) Write the basic mathematical model for Artificial neural network. [3]
e) Explain maxnet. [2]
f) Describe the concept of Self organization features maps. [3]
g) Define Fuzzification. [2]
h) Write a short notes on Fuzzy relations. [3]
i) Explain Fuzzy states. [2]
j) Illustrate about internet search technique. [3]

PART-B

(50 Marks)

2. Explain constraint satisfaction problem by taking 8 queens problem as an example. [10]

OR

3. Explain unification algorithm used for reasoning under predicate logic with an example. [10]

4. Compare and contrast between Artificial and Biological Neural Networks. [10]

OR

5. Discuss algorithm for storage of confirmation in Hop field network. Explain recall algorithm. [10]

6. Explain in detail about the concept of Fixed Weight competitive nets. [10]

OR

7. Explain the architectures of Counter-Propagation Networks and their training algorithms. [10]

8. Differentiate fuzzy set from classical set and name the properties of classical (crisp) sets. [10]

OR

9. Explain in detail about Tolerance and equivalence relations in Fuzzy sets. [10]

10. With an example show how genetic algorithms are used to build fuzzy logic control systems. [10]

OR

11. Give an example to explain how approximate reasoning is performed in Fuzzy decision making. [10]

---ooOoo—

JNTUH USED 02-12-17PM set